Project Report

Introduction and objectives for your project:

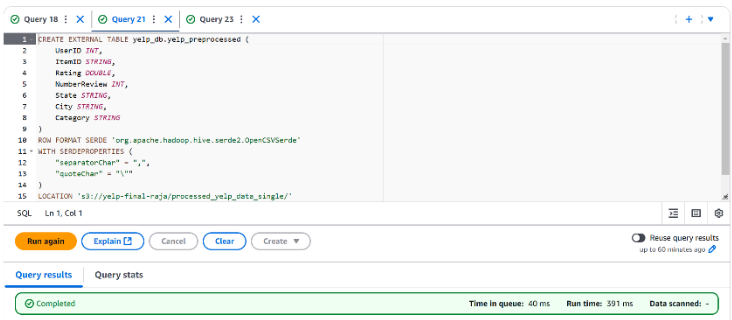
- The objective of this project is to design and implement a comprehensive data processing and analysis pipeline for a large dataset using distributed computing and big data technologies.
- The goal was to clean, transform, and analyze the data to derive actionable insights and build a recommendation system. Key technologies used included Apache Spark, AWS Athena, and AWS S3.

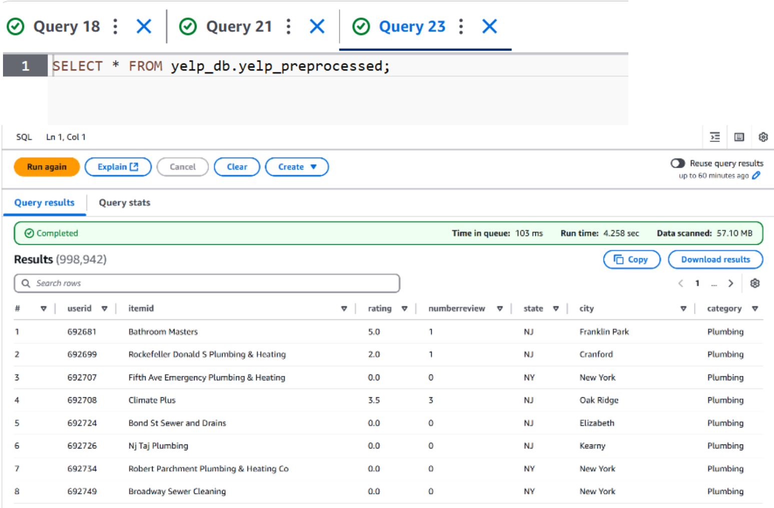
Data source:

- **Dataset Chosen**: Yelp Business Reviews Dataset (https://www.kaggle.com/datasets/abdulmajid115/yelp-dataset-contains-1-million-rows).
- **Dataset Size**: Over 1 million rows, approximately 105MB.
- **Reasoning**: The dataset's variety and volume made it suitable for distributed processing, transformation, and machine learning tasks. It also provides opportunities to analyze user reviews and ratings for building a recommendation system.

Data processing and transformation:

- Loaded raw data into AWS S3 bucket named yelp-final-raja
- Created an EMR cluster with 1 Primary and 4 Core nodes.
- Attached the cluster to a notebook and launched the cluster.
- Created a Data processing and transformation.ipynb
- Removed duplicates and null values in critical columns (e.g., ID, Organization, Rating).
- Selected and renamed relevant columns for our recommendation system.
- Aggregated data to prepare it for analysis and stored the processed data back in AWS S3.
- Used Athena to Query the data into a databases and stored them into a table.
- Tools Used: Apache Spark (PySpark), AWS S3, AWS Athena.





#	userid	itemid	rating	numberreview	state	city	category
1	692681	Bathroom Masters	5.0	1	NJ	Franklin Park	Plumbing
2	692699	Rockefeller Donald S Plumbing & Heating	2.0	1	NJ	Cranford	Plumbing
3	692707	Fifth Ave Emergency Plumbing & Heating	0.0	0	NY	New York	Plumbing
4	692708	Climate Plus	3.5	3	NJ	Oak Ridge	Plumbing
5	692724	Bond St Sewer and Drains	0.0	0	NJ	Elizabeth	Plumbing
6	692726	Nj Taj Plumbing	0.0	0	NJ	Kearny	Plumbing
7	692734	Robert Parchment Plumbing & Heating Co	0.0	0	NY	New York	Plumbing
8	692749	Broadway Sewer Cleaning	0.0	0	NY	New York	Plumbing

Machine learning model development:

Pipeline Design:

- Tools Used: Apache Spark for processing, AWS S3 for data storage, and AWS Athena for querying.
- Clustering Algorithm: Implemented KMeans Clustering using Spark MLlib to group businesses based on customer reviews.
- Feature Selection: Extracted Rating and Number of Reviews as input features for clustering.
- Pipeline Steps:
  - Preprocessed the dataset to remove duplicates and null values.
  - Transformed data into a feature vector using VectorAssembler.
  - Trained a KMeans model with 10 clusters, optimized for scalability in distributed computing.
- Orchestration: Leveraged AWS EMR for distributed processing and Athena to query intermediate results, enabling efficient pipeline execution.

Evaluation and results:

Clustering Insights:

- Analyzed **cluster sizes** to identify data distribution. Cluster 1 dominated with 73.5% of data, indicating a need for better data segmentation.
- Visualized data distribution with both bar charts and pie charts.

Recommendation Results:

- Identified high-rated businesses for specific users, such as **"Crisp"** and **"Eataly Chicago"** in Chicago for Delivery category.
- Recommendations were generated based on user's cluster, city, and category

```
VBox()
FloatProgress(value=0.0, bar_style='info', description='Progress:', layout=Layout(height
='25px', width='50%'),…
+-------------------------+------+------------+-------+--------+
|ItemID                   |Rating|NumberReview|City   |Category|
+-------------------------+------+------------+-------+--------+
|Crisp                    |4.5   |3228        |Chicago|Delivery|
|Eataly Chicago           |4.0   |3749        |Chicago|Delivery|
|Giordano's               |4.0   |2790        |Chicago|Delivery|
|Piece Brewery and Pizzeria|4.0  |3560        |Chicago|Delivery|
|Sunda Chicago            |4.0   |2835        |Chicago|Delivery|
|Xoco                     |4.0   |3646        |Chicago|Delivery|
+-------------------------+------+------------+-------+--------+
```

Metrics:

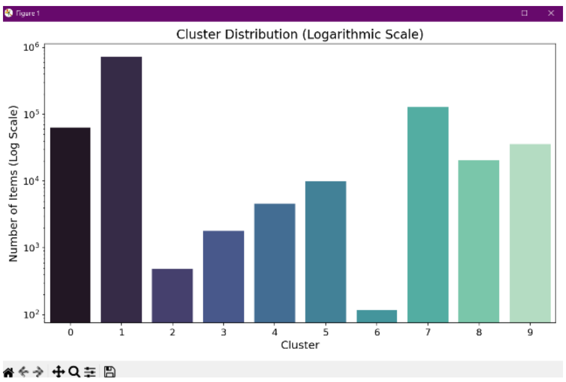
- **Model Quality**: KMeans effectively segmented businesses based on reviews and ratings.
- **User-Specific Insights**: Delivered actionable recommendations aligned with user preferences.

```
VBox()
FloatProgress(value=0.0, bar_style='info', description='Progress:', layout=Layout(height
='25px', width='50%'),…
+------+------------------+------+------------+-----+---------+--------+
|UserID|            ItemID|Rating|NumberReview|State|     City|Category|
+------+------------------+------+------------+-----+---------+--------+
|   124|       Dragon City|   3.5|          39|   AL|   Daphne|Delivery|
|   171|    Janino's Pizza|   3.5|          30|   AL|   Daphne|Delivery|
|   179|Roll & Go Sushi A...|   4.0|          10|   AL|   Daphne|Delivery|
|   187|     Marco's Pizza|   3.0|          25|   FL|Pensacola|Delivery|
|   196|Santino's Pizza &...|   4.0|           5|   FL|   Milton|Delivery|
|   220|               KFC|   3.0|           4|   AL| Saraland|Delivery|
|   246|  Godfather's Pizza|   4.0|          17|   AL|   Mobile|Delivery|
|   270|     Domino's Pizza|   2.0|          10|   AL|   Daphne|Delivery|
|   285|Hungry Howie's Pizza|   3.5|          11|   AL|   Mobile|Delivery|
|   291|      Shang Hai II|   2.5|          23|   FL|Pensacola|Delivery|
+------+------------------+------+------------+-----+---------+--------+
only showing top 10 rows
```

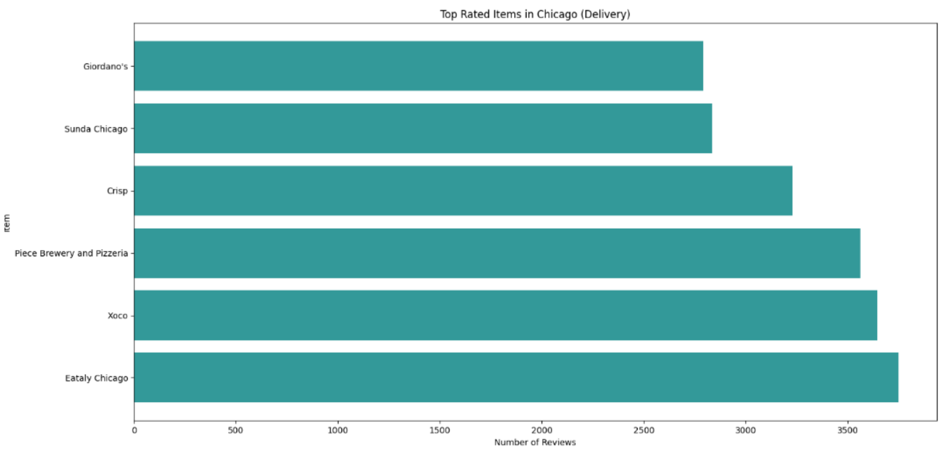
Visualizations and insights

Cluster Distribution:

- The majority of items (73.5%) are in Cluster 1, highlighting a common group of similar features.
- Smaller clusters capture niche categories or unique patterns in the data.

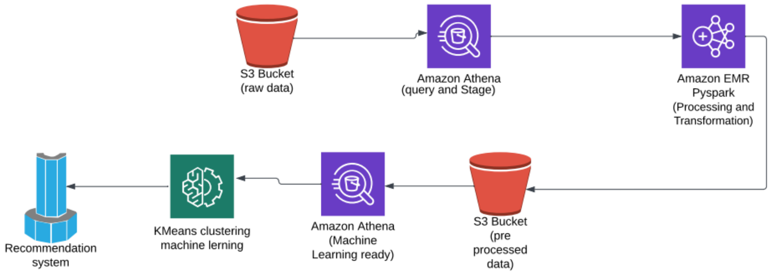


Top 6 Items recommended and their reviews for the given user:

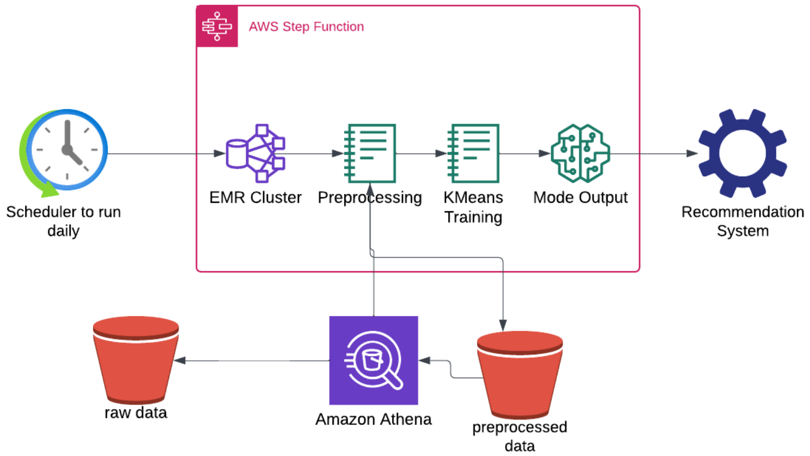


Architecture Diagrams

Current Solution



Automated Pipeline



Challenges faced:

- Processing a large dataset required for a more careful optimization and resource management.
- Integrating AWS services like S3, Athena, and EMR posed configuration challenges.
- Clustering imbalance in KMeans needed tuning for better results.
- Visualization issues in EMR led to creating plots outside the cluster.

Conclusion and future work:

- **Conclusion:** The project successfully processed large data and provided meaningful recommendations using distributed computing.
- **Future Work:** Automate the pipeline, improve clustering, and explore additional algorithms for better insights.